

CHAPTER 2



Photo of the American Hardly Gully being as

PRODUCTION POSSIBILITIES FRONTIER FRAMEWORK



In this Lecture

- An Economy's Production Possibilities Frontier
- Increasing and Constant Opportunity Costs
- The Production Possibilities Frontier and Various Economic Concepts
- Specialization, Trade and the PPF

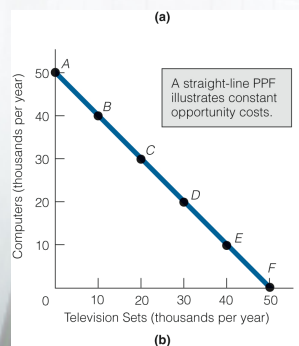


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Production Possibilities Frontier

Represents the possible combinations of two goods that can be produced in a certain period of time under the conditions of a given state of technology and fully employed resources.

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- Possible Combination to produce two product in a given time ... Fully labor and ~~the~~ the best technology

- Fully use labor

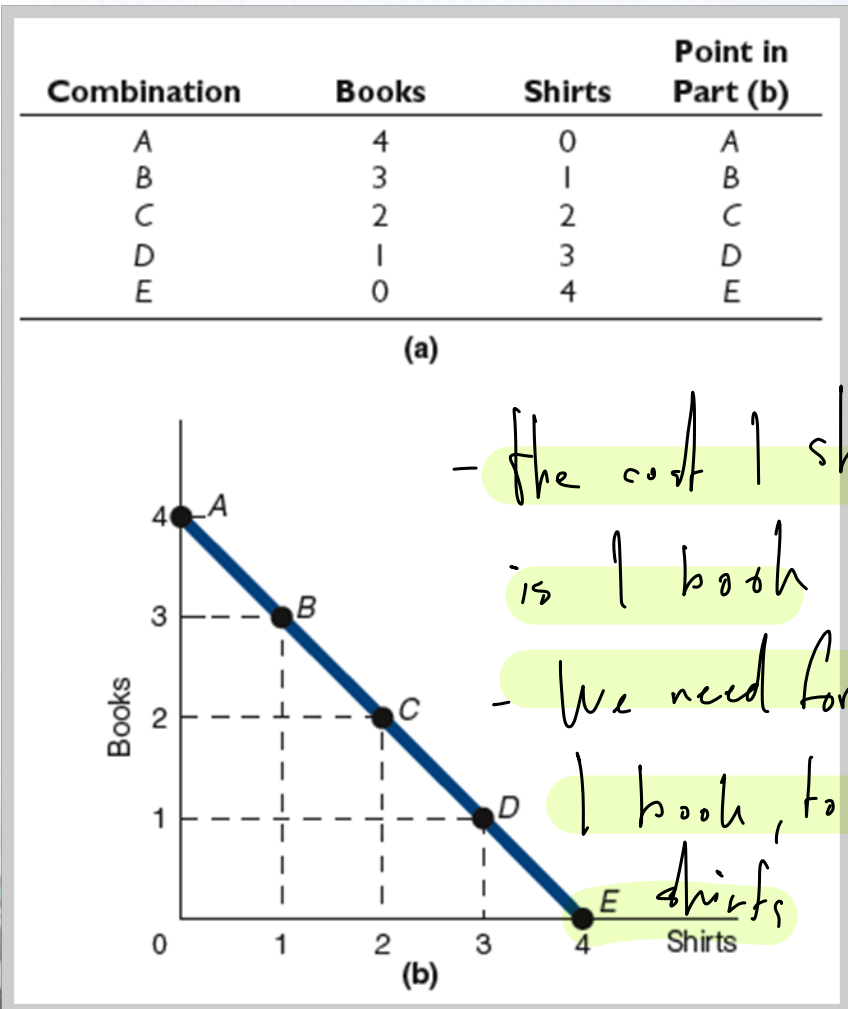
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Production Possibilities Frontier - constant

Constant Opportunity Costs

- opportunity cost

- The economy can produce any of the five combinations of books and shirts in part (a).
- We have plotted these combinations in part (b). The production possibilities frontier (PPF) in part (b) is a straight line because the opportunity cost of producing either good is constant.



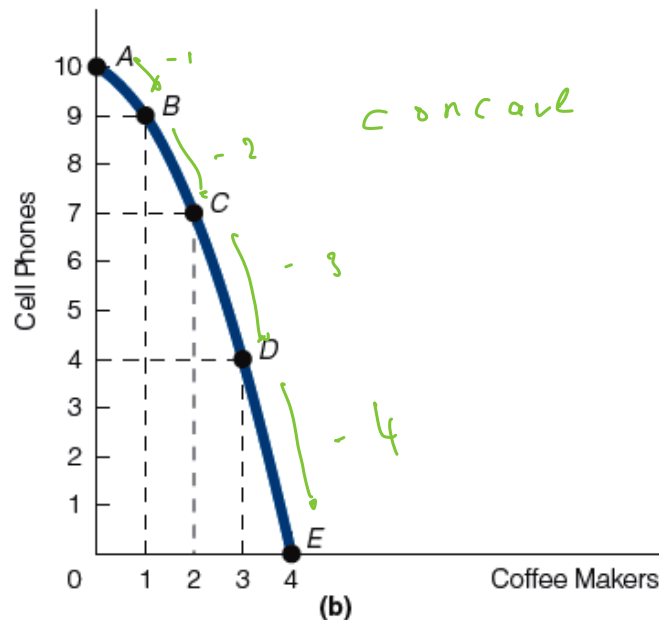
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Production Possibilities Frontier

Increasing Opportunity Costs

Combination	Cell Phones	Coffee makers	Point in Part (b)
A	10	0	A
B	9	1	B
C	7	2	C
D	4	3	D
E	0	4	E

(a)



(b)

- The economy can produce any of the five combinations of cell phones and coffee makers in part (a).
- We have plotted these combinations in part (b).
- The production possibilities frontier in part (b) is bowed outward because the opportunity cost of producing coffee makers increases as more coffee makers are produced.

1 -> 2 -> 3 -> 4
opportunity cost of making 1 unit of coffee maker increases as more coffee makers are produced

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Law of Increasing Opportunity Costs

As more of a good is produced, the opportunity costs of producing that good increase

↑ Q, ↑ OC



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Increasing Opportunity Costs

We start with the assumption that not all people can build houses at the same opportunity cost.

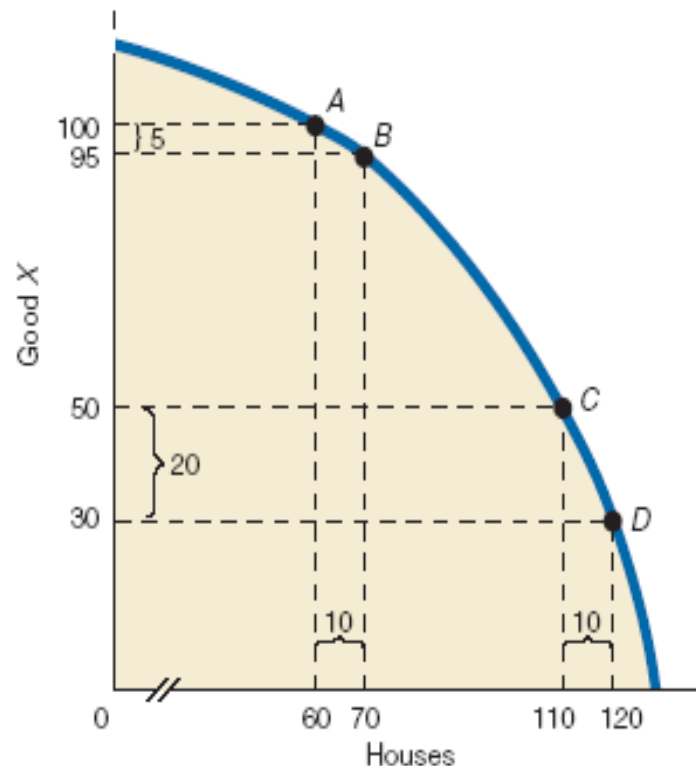
When houses are first built, only the people who can build them at (relatively) low opportunity costs will build them.

As increasingly more houses are built, people with higher opportunity costs of building houses will start building houses.

This is the same as saying that as more houses are built, the opportunity cost of building houses increases.

And this is why the PPF for houses and good X is bowed outward (concave downward). See diagram at left.

Notice that when we go from building 60 to 70 houses (10 more houses), we forfeit 5 units of good X; but when we go from building 110 to 120 houses (again, 10 more houses), we forfeit 20 units of good X.



Production Possibility Frontier Framework for Understanding



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Productive Efficiency and Inefficiency

Productive Efficiency

The condition where the maximum output is produced with given resources and technology

Productive Inefficiency

The condition where less than the maximum output is produced with given resources and technology. Productive inefficiency implies that more of one good can be produced without any less of another good being produced.

max technology
max labour use

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Technology



- Body of skills and knowledge that involve in production and known as resource

- advance: - hole produce more with a fixed resource

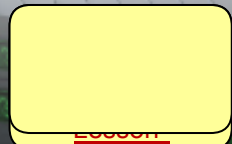
- produce same output with a fewer resource



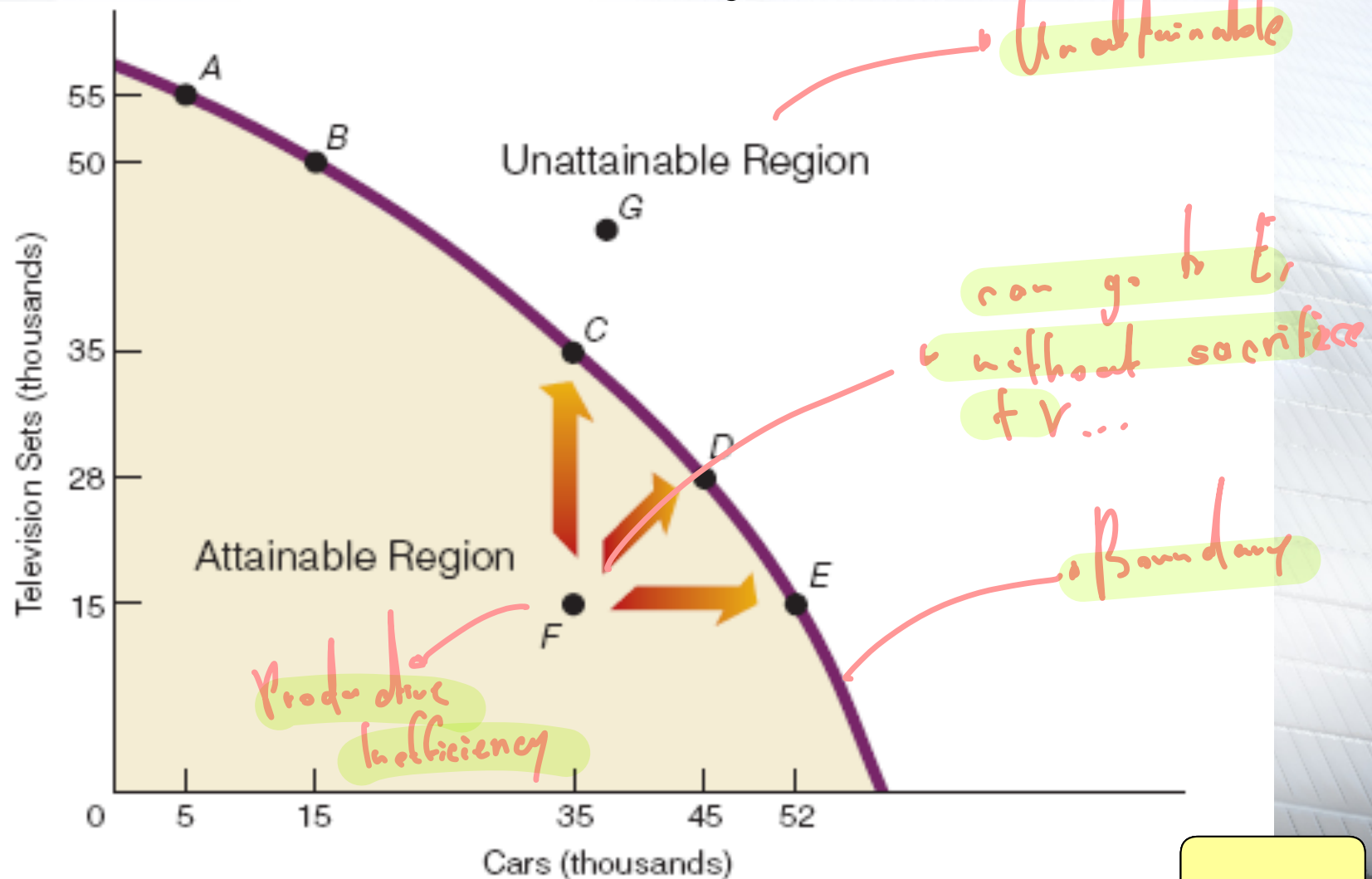
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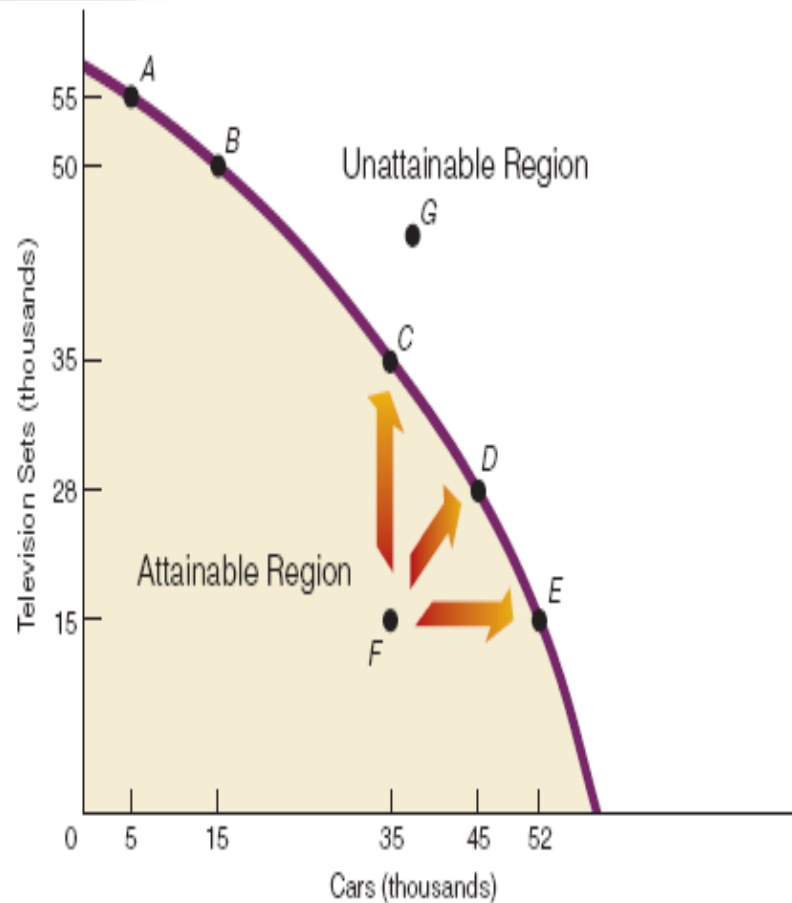
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Attainable and Unattainable Regions and Productive Efficiency



The PPF and Various Economic Concepts



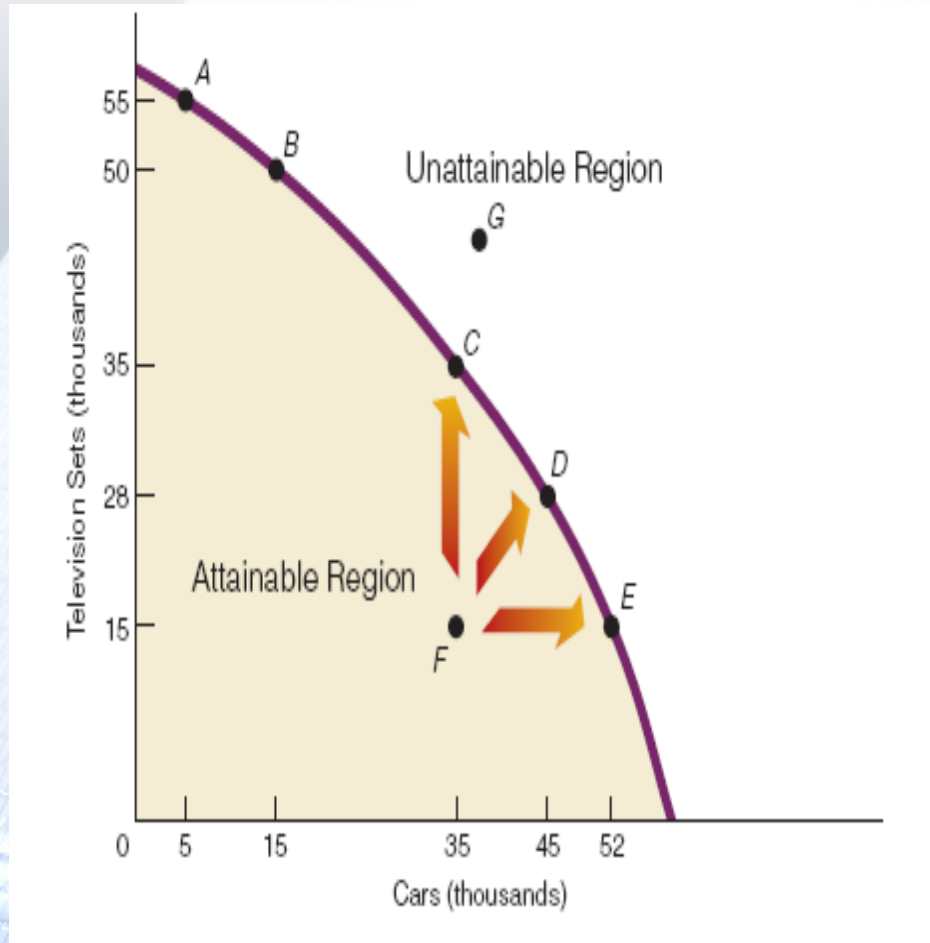
(1) **Scarcity** is illustrated by the frontier itself. Implicit in the concept of scarcity is the idea that we can have some things but not all things. The PPF separates an attainable region from an unattainable region.

(2) **Choice** is represented by our having to decide among the many attainable combinations of the two goods. For example, will we choose the combination of goods represented by point *A* or by point *B*?

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The PPF and various Economic Concepts

II

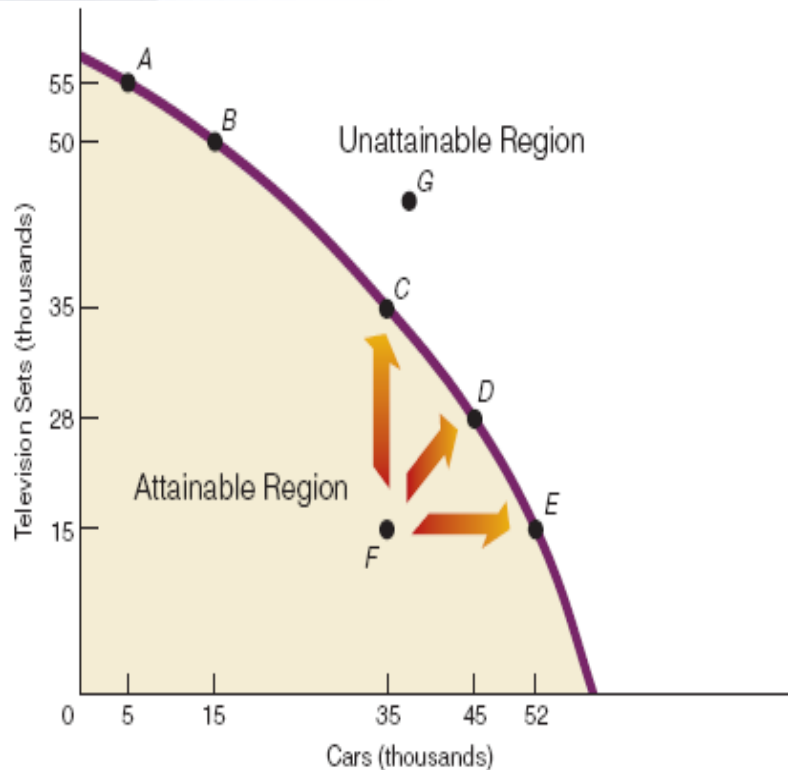


(3) **Opportunity cost** is most easily seen as movement from one point to another, such as movement from point A to point B. More cars are available at point B than at point A, but fewer television sets are available. In short, the opportunity cost of more cars is fewer television sets.

(4) **Productive efficiency** is represented by the points on the PPF (such as AE), while productive inefficiency is represented by any point below the PPF (such as F).

The PPF and various Economic Concepts

III

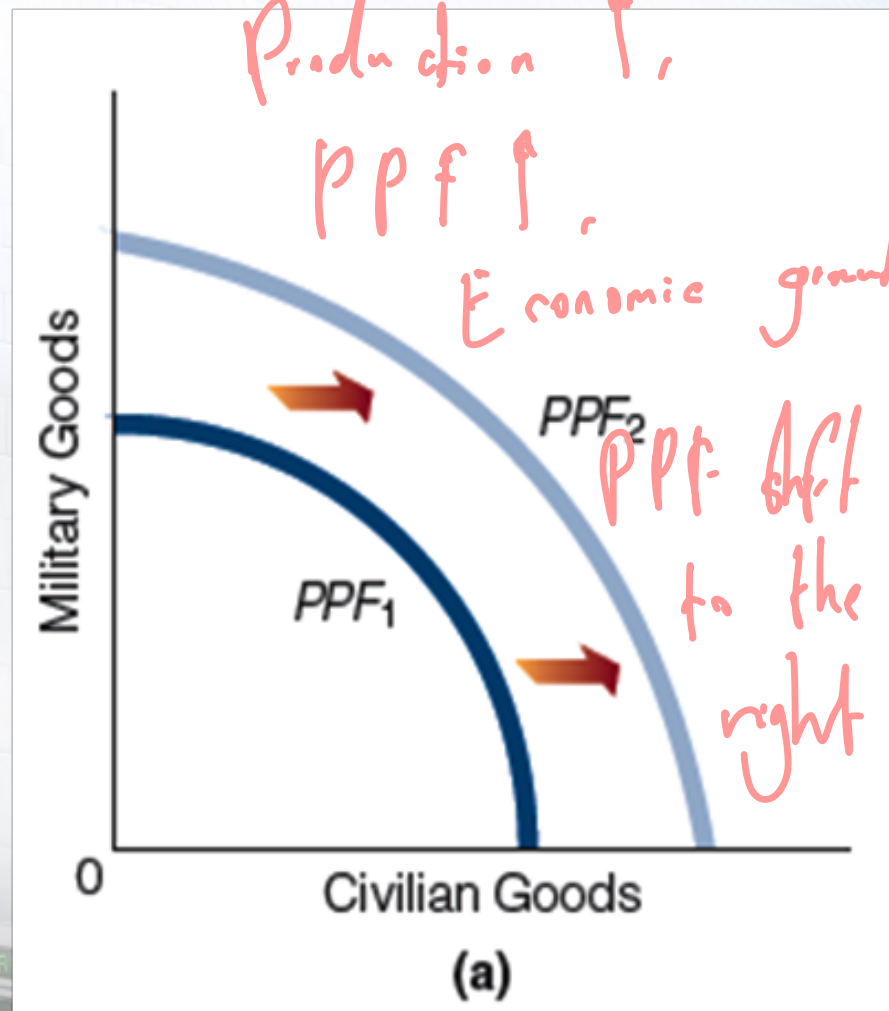


(5) *Unemployment* (in terms of resources being unemployed) exists at any productive inefficient point (such as *F*), whereas resources are fully employed at any productive efficient point (such as *AE*).

A campaign

Economic Growth within a PPF Framework

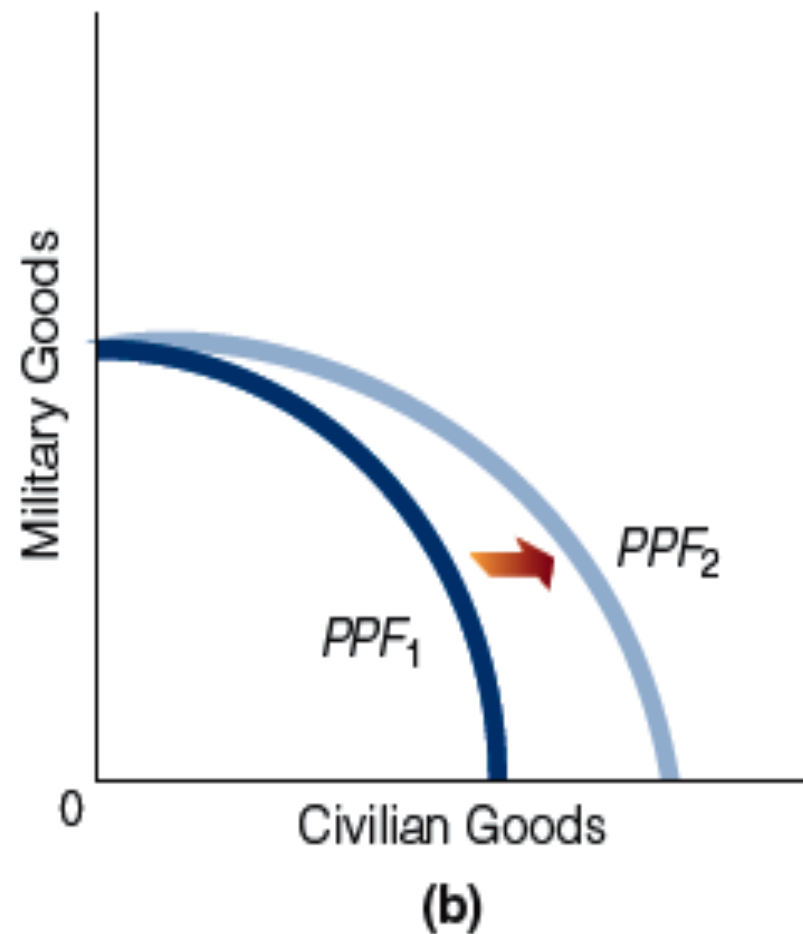
An **increase in resources** or an **advance in technology** can increase the production capabilities of an economy, leading to **economic growth** and shift outward in the production possibilities frontier.



Bole ↑ production
with 2 product

Economic Growth within a PPF Framework

If the advance in technology leads to the greater production of only one good (such as civilian goods in our exhibit), then the PPF shifts outward, as shown in (b).



Blue ↑ production
military ↑ civilian product

Exchange or Trade

- Between two parties
- to reduce the cost of production, we can specialized some item
- firm A can produce banyak barang dgn opportunity cost yg

Comparative Advantage

The situation where someone can produce a good at lower **opportunity cost** than someone else can.

lebih rendah
berbanding firm B

Specialization

Producing goods in which you have a **comparative advantage**

• kita specialize utk kurangkan kos satu product untuk produksi

• comparative advantage berbeza dgn absolute advantage

A Timeline for Trade

Ex Ante - Phrase that means before,” as in before a trade.

Ex Post - Phrase that means “after,” as in after a trade.



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Ex Ante

constant = straight line

Elizabeth and Brian Before the Trade

- ada comparative advantage
- schak opportunity cost di
rendah
- so dia specialize
bread

Elizabeth

Bread

Apples

20

0

10

10

20

Brian

Bread

Apples

10

0

5

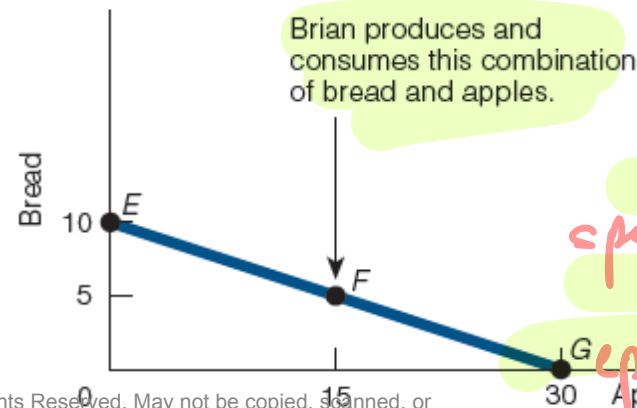
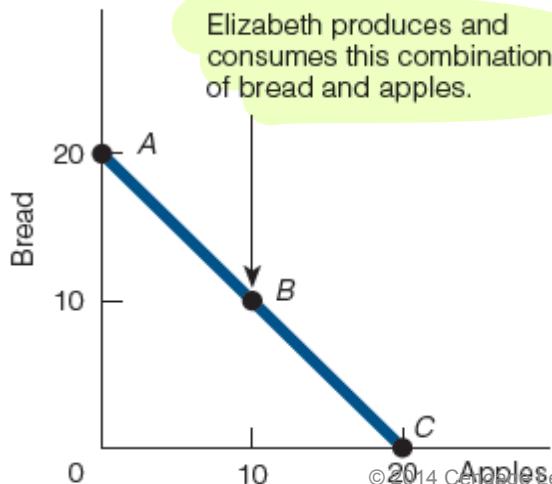
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0

30

1 bread = 3 apples

- ada comparative advantage
dia produce apple ...



so dia specialize
pengeluar apel

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Ex Post

Elizabeth bagi 8 Bread
Brian bagi 12 apple
] ini conth ayat

		No Specialization and No Trade	Specialization and Trade	Gains from Specialization and Trade
Elizabeth	Consumption of Loaves of Bread	10	12	+2
	Consumption of Apples	10	12	+2
Brian	Consumption of Loaves of Bread	5	8	+3
	Consumption of Apples	15	18	+3

